Cyclohexane Reactor Ver 1 · operator training simulator

Electrical load list

Document no. CHX-V1-EE-LL-001 · **Discipline** Electrical engineering

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

- 8 motors (5 running, 3 standby); absorbed power from the simulator model (econ() terms, JI-101; 120 % rate from owRun).
- Normal running 1 741 kW; maximum demand 2 020 kVA at design, 2 298 kVA at 120 % rate.
- LV transformer 2 × 100 % 630 kVA (6.6/0.42 kV); MV motors K-101 and M-101 on the site 6.6 kV system; UPS 2 × 10 kVA.
- Every value not taken from the model is labelled assumed or typical practice (section 12).

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Basis

Item	Basis
Scope	Battery limit of the reactor train: R-101 … V-104, K-101, D-101 and their pumps. Utilities, tank farm, flare and control room are outside.
Supply	Site 6.6 kV, 3-phase, 50 Hz (assumed); unit LV switchboard 400/230 V fed by 6.6/0.42 kV transformers.
Voltage level	LV 400 V for motors up to 200 kW; MV 6.6 kV above (typical practice; company standards put the break between 150 and 300 kW).
Absorbed power	From the model at design (noise off): each motor's econ() term (motor switched off on a copy, difference of econ().kW) and JI-101; 120 % rate from owRun (FIC-101 41.76 t/h, hydrogen in ratio).
Rating case	Largest of design, 120 % rate and (agitator) ungassed; the motor is also never smaller than the model's maximum.
Margins	Pumps ×1.25 below 22 kW, ×1.15 for 22–55 kW, ×1.10 above (API 610 style); compressor ×1.10 (API 617 style); agitator ×1.10 over ungassed. Typical practice.
Motor sizes	IEC standard output series (… 30, 37, 45, 55, 75, 90, 110, 132, 160, 200, 250, 315 … 1400, 1600 kW).
Efficiency, power factor	Typical full-load values of IE3 4-pole motors by size: ≤ 37 kW 0.93 / 0.85; ≤ 110 kW 0.95 / 0.86; ≤ 355 kW 0.96 / 0.87; ≤ 1000 kW 0.965 / 0.88; larger 0.97 / 0.89.
Maximum demand	1.0 × continuous + 0.3 × intermittent + 0.1 × standby (diversity factors assumed, typical practice); kW and kvar summed separately.
Growth	20 % on transformer and UPS sizes (assumed).
Cables	Cu, XLPE, multicore, in air on perforated trays; ampacity of IEC 60364-5-52 method E style (typical values); ambient 40 °C ×0.91, 3 circuits touching ×0.82; ΔV ≤ 5 % running, ≤ 15 % starting; MV minimum 70 mm² for short-circuit withstand (assumed).
Motor starting	DOL locked-rotor current 6.5 × FLC at pf 0.3, locked-rotor torque 0.8 p.u. (typical); dip limit 10 % at the board (typical); site 6.6 kV fault level 125 MVA, transformer impedance 6 % (assumed).

1.1 · Typical motor efficiency and power factor (full load, IE3 4-pole)

Rated output up to kW	η	pf
37	0.930	0.85
110	0.950	0.86
355	0.960	0.87
1 000	0.965	0.88
larger	0.970	0.89

1.2 · Consumers outside the battery limit (not in this list)

Cooling-water pumps and cooling-tower fans (E-103, E-106, R-102 jacket in COOL) · Boiler feed-water pumps (D-101) · Instrument-air and nitrogen compressors · Hydrogen make-up supply (hydrogen plant) · Tank-farm pumps: benzene supply to TK-100, product from TK-106 / TK-107 · Fire-water pumps · Flare FL-100 and its ignition panel · Control-room building services

2 · Load list at design

Absorbed kW from the model at design; rating case = largest of design, 120 % and (agitator) ungassed; motor kW = next IEC size ≥ max(margin × rating case, model maximum). Input kW = absorbed ÷ η; kVA = input ÷ pf; FLC at the rated output.

No	Tag	Service	Duty	Abs. kW	Rating case kW	Margin	Motor kW	Voltage	Load factor	η	pf	Input kW	kVA	FLC A	Note
1	P-100A	Benzene feed pump (TK-100 → FIC-101 → R-101)	continuous	90.0	90.0	1.10	110	LV 400 V	0.82	0.950	0.86	94.7	110.2	194	econ() constant: does not change with rate
2	P-100B	Benzene feed pump (TK-100 → FIC-101 → R-101) (spare)	standby	90.0	90.0	1.10	110	LV 400 V	0.82	0.950	0.86	94.7	110.2	194	standby; separate bus section from P-100A
3	P-101A	R-101 pump-around pump (through E-101 / FV-104)	continuous	160.0	160.0	1.10	200	LV 400 V	0.80	0.960	0.87	166.7	191.6	346	econ() constant; VSD option studied in the Efficiency tab
4	P-101B	R-101 pump-around pump (through E-101 / FV-104) (spare)	standby	160.0	160.0	1.10	200	LV 400 V	0.80	0.960	0.87	166.7	191.6	346	standby; auto-start recommended (Reliability tab); separate bus section
5	P-102A	R-102 transfer pump (through F-102 to R-103)	continuous	30.0	30.0	1.15	37	LV 400 V	0.81	0.930	0.85	32.3	38.0	68	econ() constant
6	P-102B	R-102 transfer pump (through F-102 to R-103) (spare)	standby	30.0	30.0	1.15	37	LV 400 V	0.81	0.930	0.85	32.3	38.0	68	standby; separate bus section
7	K-101	Recycle hydrogen compressor (V-104 → FIC-103 → R-101 sparger)	continuous	1 200.2	1 440.2	1.10	1 600	MV 6.6 kV	0.75	0.970	0.89	1 237.3	1 390.2	162	econ(): 1200 kW × Q/Qd (0.5–1.3); type not fixed by the model, started unloaded (assumed); no spare
8	M-101	R-101 agitator (Raney-Ni slurry, gas dispersion)	continuous	153.0	180.0	1.10	250	MV 6.6 kV	0.61	0.960	0.87	159.3	183.2	26	gassed at design (JI-101); rated on the model maximum (ungassed, $n_1 \geq 1.3\ n_1 d$); VSD for a settled-slurry start
9	UPS-100	UPS rectifier input (instrument, SIS, DCS, F&G, telecom)	continuous	–	–	–	–	LV 400 V	–	–	0.90	7.7	8.6	–	computed from the UPS list (part 6); its loads are assumed
10	LT-100	Normal lighting (area, platforms, stairs)	continuous	–	–	–	–	LV 400 V	–	–	0.95	12.0	12.6	–	assumed allowance; emergency luminaires self-contained (3 h battery, assumed)
11	AH-100	Analyzer shelter HVAC and services	continuous	–	–	–	–	LV 400 V	–	–	0.85	8.0	9.4	–	assumed allowance; shelter ventilation keeps it non-hazardous inside or Zone 2 (HAC-001)
12	HT-100	Electric heat tracing, analyzer sample lines	continuous	–	–	–	–	LV 400 V	–	–	1.00	6.0	6.0	–	assumed allowance; no freeze protection needed in the Thai climate (benzene melts at 5.5 °C)
13	K-101-AUX	K-101 auxiliaries (lube / seal oil pumps, coolers, motor space heater)	continuous	–	–	–	–	LV 400 V	–	–	0.85	15.0	17.6	–	assumed allowance; type of K-101 not fixed by the model
14	M-101-AUX	M-101 seal support unit (barrier-fluid pump and cooler)	continuous	–	–	–	–	LV 400 V	–	–	0.85	2.2	2.6	–	assumed allowance; dual pressurised seal (assumed)
15	WS-100	Maintenance and welding sockets	intermittent	–	–	–	–	LV 400 V	–	–	0.80	40.0	50.0	–	assumed allowance; used during maintenance and catalyst change-out

No	Tag	Service	Duty	Abs. kW	Rating case kW	Margin	Motor kW	Voltage	Load factor	η	pf	Input kW	kVA	FLC A	Note
16	HO-100	Hoists (catalyst handling on the R-101 top, F-102 change-out)	intermittent	–	–	–	–	LV 400 V	–	–	0.80	7.5	9.4	–	assumed allowance;
–	E-101	Pump-around cooler / LP steam generator	none												no electrical load: no drive; MP steam (HV-110) for start-up heating is a utility
–	D-101	Steam drum	none												no electrical load: no drive; BFW pumps are outside the battery limit
–	E-103, E-106	Reflux condenser, product cooler (cooling water)	none												no electrical load: no drive; CW pumps outside the battery limit
–	R-102, R-103, V-103, V-104, F-102A/B	Reactors, drums and catalyst filters	none												no electrical load: no drive (R-102 jacket on MP steam or CW)

3 · Totals and maximum demand

Maximum demand = 1 × continuous + 0.3 × intermittent + 0.1 × standby (diversity factors assumed, typical practice); kW and kvar summed separately. 120 % rate: K-101 and M-101 from owRun (hydrogen in ratio); the pump powers of econ() are constants. Spares running: every standby pump counted as running (change-over).

kW unless stated	Design	120 % rate	Spares running
Normal running (continuous), kW	1 741	1 989	2 035
Normal running, kVA	1 968	2 247	2 308
Standby, kW	294	294	0
Intermittent, kW	48	48	48
Maximum demand, kW	1 785	2 032	2 049
Maximum demand, kVA	2 020	2 298	2 325
Maximum demand, pf	0.88	0.88	0.88
of which LV 400 V, kVA	447	447	753
of which MV 6.6 kV, kVA	1 573	1 851	1 573
Motor absorbed power running, kW	1 633	1 873	1 913
LV transformer chosen, kVA	630	630	1 000

4 · Transformer and supply

LV transformer 6.6/0.42 kV: LV maximum demand 447 kVA × (1 + 20 % growth, assumed) = 536 kVA → standard size 630 kVA, 2 × 100 % (each able to carry the whole board, secondary-selective with a bus-tie: typical practice for a continuous unit); loading at maximum demand 71 %. At 120 % rate the LV demand is 447 kVA (transformer 630 kVA still enough). The unit's total demand on the site 6.6 kV system is 2 020 kVA (2 424 kVA with growth); the 6.6 kV supply and its transformers are outside the battery limit. A and B motors of each pair are on different LV bus sections (EQ-ee-7).

5 · Cross-check of the model powers

Independent estimate: pumps flow × pressure rise ÷ η (P-100 Δp 35 bar from tank to R-101, assumed; P-101 and P-102 the model's 3 bar pump Δp); K-101 ideal-gas compression of the model flow and pressure ratio with k 1.4 and η 0.75 (assumed). A model value well above the estimate is conservative for the motor but inflates demand and energy: raised as an engineering query (EQ-ee-1, EQ-ee-3).

Machine	Flow	Δp bar	η assumed	Estimate kW	Model kW	Model ÷ estimate	Model ≥ estimate
P-100	40.6 m³/h	35.00	0.6	65.8	90.0	1.4	✓
P-101	589.7 m³/h	3.00	0.75	65.5	160.0	2.4 → EQ	✓
P-102	32.6 m³/h	3.00	0.6	4.5	30.0	6.6 → EQ	✓
K-101	46 000 Nm³/h	3.00 (ratio 1.100)	0.75	194.1	1 200.2	6.2 → EQ	✓

6 · Motor feeder cables

Cu XLPE multicore in air on perforated trays; ampacity IEC 60364-5-52 method E style (typical values); derating ambient 40 °C × 0.91, 3 circuits touching × 0.82; ΔV = √3·I·L·(R cos φ + X sin φ) with R at 90 °C, X 0.08 Ω/km (LV) and 0.1 Ω/km (MV); limits 5 % running and 15 % starting (starting current: DOL 6.5, soft starter 3.5, VSD 1.1 × FLC at pf 0.3); MV minimum 70 mm² for the short-circuit duty (assumed).

Motor	Output kW	Voltage	FLC A	Length m (assumed)	Starting	Size by ampacity mm²	Chosen mm²	ΔV running %	ΔV starting %
P-100A	110	400 V	194	120	DOL	95	95	2.56	9.9
P-100B	110	400 V	194	120	DOL	95	95	2.56	9.9
P-101A	200	400 V	346	90	Soft starter	240	240	1.68	5.0
P-101B	200	400 V	346	90	Soft starter	240	240	1.68	5.0
P-102A	37	400 V	68	100	DOL	16	16	3.78	9.8
P-102B	37	400 V	68	100	DOL	16	16	3.78	9.8
K-101	1 600	6.6 kV	162	70	DOL	70	70	0.10	0.4
M-101	250	6.6 kV	26	90	VSD	4	70	0.02	0.0

6.1 · Cable data used (typical)

mm²	2.5	4	6	10	16	25	35	50	70	95	120	150	185	240	300
Ampacity A	32	42	54	75	100	127	158	192	246	298	346	399	456	538	621
R Ω/km	9.45	5.88	3.93	2.33	1.47	0.927	0.668	0.494	0.342	0.247	0.196	0.159	0.128	0.0977	0.0787

7 · Motor-starting study

Dip at the board ≈ S_start / (S_sc + S_start). Torque available = locked-rotor torque 0.8 p.u. × (I/I_LR)² (soft starter) or × tap² (autotransformer) × (1 − dip)²; a VSD gives 1 p.u. Required ≥ 1.2 × load breakaway torque (compressor started unloaded 0.2, agitator in settled slurry 0.6, centrifugal pump 0.15 p.u.; all assumed). Dip limit 10 % (typical). Site 6.6 kV fault level 125 MVA (assumed; weak case 60 MVA); LV board from the 630 kVA transformer with 6 % impedance.

7.1 · K-101 1600 kW 6.6 kV, source 125.0 MVA

Method	I start ÷ FLC	I start A	Starting kVA	Dip %	Torque p.u.	Dip ≤ 10 %	Starts the load
Direct on line (DOL)	6.50	1 054	12 047	8.8	0.67	✓	✓
Soft starter (current limit 3.5 × FLC)	3.50	567	6 487	4.9	0.21	✓	✗
Autotransformer, 65 % tap	2.75	446	5 097	3.9	0.31	✓	✓
Variable-speed drive (VSD)	1.10	178	2 039	1.6	1.00	✓	✓

7.1b · K-101 1600 kW 6.6 kV, source 60.0 MVA · weak supply

Method	I start ÷ FLC	I start A	Starting kVA	Dip %	Torque p.u.	Dip ≤ 10 %	Starts the load
Direct on line (DOL)	6.50	1 054	12 047	16.7	0.55	✗	✓
Soft starter (current limit 3.5 × FLC)	3.50	567	6 487	9.8	0.19	✓	✗
Autotransformer, 65 % tap	2.75	446	5 097	7.8	0.29	✓	✓

Method	I start ÷ FLC	I start A	Starting kVA	Dip %	Torque p.u.	Dip ≤ 10 %	Starts the load
Variable-speed drive (VSD)	1.10	178	2 039	3.3	1.00	✓	✓

7.2 · M-101 250 kW 6.6 kV, source 125.0 MVA

Method	I start ÷ FLC	I start A	Starting kVA	Dip %	Torque p.u.	Dip ≤ 10 %	Starts the load
Direct on line (DOL)	6.50	170	1 946	1.5	0.78	✓	✓
Soft starter (current limit 3.5 × FLC)	3.50	92	1 048	0.8	0.23	✓	✗
Autotransformer, 65 % tap	2.75	72	823	0.7	0.33	✓	✗
Variable-speed drive (VSD)	1.10	29	329	0.3	1.00	✓	✓

7.2b · M-101 250 kW 6.6 kV, source 60.0 MVA · weak supply

Method	I start ÷ FLC	I start A	Starting kVA	Dip %	Torque p.u.	Dip ≤ 10 %	Starts the load
Direct on line (DOL)	6.50	170	1 946	3.1	0.75	✓	✓
Soft starter (current limit 3.5 × FLC)	3.50	92	1 048	1.7	0.22	✓	✗
Autotransformer, 65 % tap	2.75	72	823	1.4	0.33	✓	✗
Variable-speed drive (VSD)	1.10	29	329	0.5	1.00	✓	✓

7.3 · P-101A 200 kW 400 V, source 10.5 MVA

Method	I start ÷ FLC	I start A	Starting kVA	Dip %	Torque p.u.	Dip ≤ 10 %	Starts the load
Direct on line (DOL)	6.50	2 247	1 557	12.9	0.61	✗	✓
Soft starter (current limit 3.5 × FLC)	3.50	1 210	838	7.4	0.20	✓	✓
Autotransformer, 65 % tap	2.75	950	659	5.9	0.30	✓	✓
Variable-speed drive (VSD)	1.10	380	263	2.4	1.00	✓	✓

Result: K-101 starts DOL on a strong 6.6 kV supply; on a weak supply an autotransformer or a VSD is needed (a soft starter cannot give the torque). M-101 is listed with a VSD (breakaway torque of a settled slurry; speed for ungassed operation). P-101A/B are listed with soft starters because DOL dips the 400 V board beyond the limit.

8 · UPS and essential loads

The SIS is de-energise to trip: losing its supply trips the unit (fail-safe). The UPS keeps the unit running through dips (availability) and keeps the first-out, the alarms, fire and gas detection and communications alive. Loads assumed.

Load	Tags	VA (assumed)	On loss of supply	Note
SIS logic solver with its I/O (6 trip inputs, ESD pushbutton, outputs, first-out)	SIS-100, HS-100	800	outputs de-energise: unit trip	de-energise to trip: losing the UPS trips the unit (fail-safe); the UPS is there for availability
SIS final elements: solenoids of the unit-trip valves	XV-101, XV-105	30	valves close	de-energise to close (fail closed)
Depressuring valve solenoid	EDV-101	15	valve opens (fail-open assumed)	opens only on ESD in the model; on loss of SIS power it would depressure the loop (EQ-ee-6)
DCS controllers and I/O for the unit, loop-powered transmitters and positioners	12 loops, 6 trip sensors, indicators	2 500	control lost; valves go to their fail positions	fail positions as in the Safety layers tab
Operator consoles (unit share)	DCS HMI	1 000	operators lose the view	control room outside the battery limit
Analyzers and sample systems	AI-100, AI-103, AI-104, AI-105	1 500	analysis lost	AI-100 feeds the hydrogen permissive; GC oven heaters on normal supply (assumed)
Fire and gas system	GD-1…3, FD-1, MCP-1 (recommended in Safety layers)	600	detection lost	must stay alive through a power failure
Telecom, PA and CCTV (unit share)	PA, radio base	800	communication lost	needed to run an emergency
Total		7 245	× (1 + 25 %) = 9.1 kVA → 2 × 10 kVA UPS (redundant, assumed)	battery 30 min at 5.8 kW → 4.0 kWh (inverter η 0.9, depth of discharge 0.8); rectifier input 7.7 kW

9 · Behaviour on loss of power (simulator model)

Design state, trips on, no operator action. Times from the model runs of the Process reliability tab (tabs/relia/rl_dyn.json); the Electrical tab re-runs the first trips in the page.

Motor(s) stopped	What the unit does	Time	Electrical design consequence
M-101 agitator	XS-101 trips the unit (agitator stopped with XV-101 open): XV-101 and XV-105 close, K-101 stops, E-101 goes to full cooling, the R-102 jacket to COOL.	10.25 s	Keep M-101 running through short dips (undervoltage release with a time delay, assumed 0.5–1 s); a restart needs the SIS reset and the start permissives (agitator before benzene).
K-101 compressor	XS-102 trips the unit (compressor stopped with XV-101 open) with the same actions.	10.25 s	MV motor: delayed undervoltage trip so dips of a few hundred ms do not stop it; restart is a planned sequence, not automatic.
P-101 running pump, spare not started	TI-100 reaches the 210 °C alarm at 1:10, the 220 °C TSHH-100 setting at 2:50 and the unit trips at 2:55: a ride-through of about 2 min 50 s. A spare started within ≈ 169 s keeps TI-100 below 220 °C; started at 5 s it peaks at 200.7 °C.	2:55	P-101A and P-101B on different LV bus sections; an auto-start of the spare (recommended in the Reliability tab) needs the spare's starter healthy and its bus alive; re-acceleration after a dip helps more than anything else here.
P-100 running pump	The benzene feed stops at once; no trip within 60 min; R-101 cools below the 140 °C light-off after 6:16.	—	Production stops but the unit stays safe; A and B on separate bus sections; no automatic restart needed.
P-102 running pump	Flow continues on the pressure difference; the R-102 level rises slowly: LIC-102 H 80 % after 54 min, no trip within 3 h.	54 min	Slow: a manual start of P-102B is enough; separate bus sections.
All motors (loss of supply)	XS-101 trips the unit first; XV-101 and XV-105 de-energise and close; FV-104 fails open but no pump-around flows, so R-101 heat leaves by boil-up to E-103 while cooling water lasts.	10.25 s	SIS, DCS and F&G on the UPS keep the trip, the first-out and the alarms. A site-wide loss also stops the CW and BFW pumps (outside the battery limit): a relief case for the Relief & vent tab.

Motor(s) stopped	What the unit does	Time	Electrical design consequence
UPS lost (instrument power)	SIS outputs de-energise: unit trip (fail-safe). Control valves go to their fail positions. EDV-101 would open if its solenoid is de-energise-to-open.	—	Two UPS in parallel with separate batteries (assumed); decide whether EDV-101 should open on loss of SIS power (EQ-ee-6).

10 · Checks against the model

Computed when this document was generated; the Electrical tab recomputes them (and more) live in the page, part 11.

Check	Values	
Every motor of the model (MOTORS) is in the list and no other	8 / 8	✓
Duty matches the design switches (running = continuous)	P-100A, P-100B, P-101A, P-101B, P-102A, P-102B, K-101, M-101	✓
Total motor absorbed power running = econ() kW at design	1 633.2 vs 1 633.2 kW (FACTS 1 633.2)	✓
M-101 design power = JI-101 at design	153.0 vs 153.0 kW	✓
P-100A rating ≥ max(margin × rating case, model maximum)	110 ≥ 99.0 kW	✓
P-100B rating ≥ max(margin × rating case, model maximum)	110 ≥ 99.0 kW	✓
P-101A rating ≥ max(margin × rating case, model maximum)	200 ≥ 176.0 kW	✓
P-101B rating ≥ max(margin × rating case, model maximum)	200 ≥ 176.0 kW	✓
P-102A rating ≥ max(margin × rating case, model maximum)	37 ≥ 34.5 kW	✓
P-102B rating ≥ max(margin × rating case, model maximum)	37 ≥ 34.5 kW	✓
K-101 rating ≥ max(margin × rating case, model maximum)	1600 ≥ 1 584.2 kW	✓
M-101 rating ≥ max(margin × rating case, model maximum)	250 ≥ 204.3 kW	✓
Voltage follows the threshold (MV above 200 kW)	P-100A LV, P-101A LV, P-102A LV, K-101 MV, M-101 MV	✓
LV transformer ≥ LV maximum demand × (1 + growth)	630 ≥ 536 kVA	✓
SIS logic solver and final elements on the UPS	SIS-100, XV-101, XV-105, EDV-101	✓
K-101 120 % power = econ() coefficient × owRun gas flow	1 440.2 kW at 55 200 Nm³/h	✓

11 · Engineering queries

ID	Query	Documents	Proposed resolution
EQ-ee-1	K-101 power in econ() is 1200.2 kW at design, about 6.2 × a gas-compression estimate (194.1 kW for 46 000 Nm³/h at a pressure ratio 1.100, k 1.4, η 0.75 assumed). The load list carries the model value, which sets a 1600 kW MV motor and most of the demand.	model econ() · LL-001 · Process engineering (compressor hydraulics) · Mechanical (K-101 datasheet)	Process engineering to state the K-101 type, rated flow and head; if the absorbed power is near the estimate, K-101 needs about a 250 kW motor and the MV board, transformer and starting study shrink. Until then keep the conservative model value.
EQ-ee-2	econ() charges the motors' absorbed (shaft) power as electricity (1633.2 kW); with typical motor efficiencies the electrical input is 1690.3 kW, plus 50.9 kW of other continuous loads.	model econ() · Efficiency tab (43.6 kWh/t) · LL-001	State in the Efficiency tab that its power is shaft power; or divide by η in econ() (lead decision: model change).
EQ-ee-3	Pump powers in econ() are constants that do not move with rate (P-100 90, P-101 160, P-102 30 kW). With the model's 3 bar pump ΔP, P-101 needs ≈ 65.5 kW and P-102 ≈ 4.5 kW (η assumed).	model econ() · LL-001 · Process engineering (pump hydraulics)	Process engineering to issue pump hydraulics (rated flow, head, NPSH); the load list keeps the larger model values until then.
EQ-ee-4	The boiling ceiling of R-101 (TbC at PI-101 ≈ 258.3 °C) is above the auto-ignition temperature of cyclohexane (≈ 245 °C, typical literature value). Only TSHH-100 at 220 °C keeps the reactor and pump-around surfaces below it.	HAC-001 · Safety layers (critical limits) · SRS (SIF-1)	Keep R-101 and pump-around insulation and cladding intact (no oil-soaked lagging); record hot-surface ignition as a consequence in the HAZOP / LOPA of the runaway scenario; electrical equipment stays T3.
EQ-ee-5	The model lets the agitator power reach 204.3 kW (ungassed, $n_1 \geq 1.3\ n_1 d$) above the 200 kW IEC step; the heat balance in step() uses 180 kW while JI-101 reads 153 kW gassed.	model step() / measure() · LL-001 · Mechanical (M-101 datasheet)	M-101 rated 250 kW until the agitator datasheet gives the maximum ungassed power at the highest level and the coldest start-up.

ID	Query	Documents	Proposed resolution
EQ-ee-6	EDV-101 is assumed fail-open and the SIS de-energises to trip: a loss of SIS power would open EDV-101 and depressure the hydrogen loop to the flare, although only the ESD opens it in the model.	Safety layers (SL_ASSUMED) · SRS · C&E · LL-001 UPS list	Decide in the SRS whether loss of SIS power is an ESD (EDV-101 opens) or a unit trip (EDV-101 energise-to-open with line monitoring, or held shut on a separate supply).
EQ-ee-7	Loss of the running P-101 leaves about 2 min 50 s before TSHH-100 (Reliability tab run). An auto-start of P-101B only helps if A and B are on separate bus sections and the bus survives the event.	Reliability tab · LL-001 · single-line diagram (not issued)	Show P-101A and P-101B (and the other A/B pairs) on different LV bus sections in the single-line diagram; add re-acceleration for dips.

12 · Assumptions

Site supply 6.6 kV, 50 Hz; LV 400/230 V · Voltage threshold 200 kW · Motor margins, IEC sizes, η and pf by size · Diversity factors 1.0 / 0.3 / 0.1 · Growth 20 % (transformer), 25 % (UPS) · Cable lengths 70–120 m · Cable ampacity, resistance and derating (typical tables) · Locked-rotor current 6.5 × FLC, torque 0.8 p.u.; load torques compressor 0.2, agitator 0.6, centrifugal pump 0.15 p.u.; soft-starter current limit 3.5 × FLC · Site 6.6 kV fault level 125 MVA (60 MVA weak case); transformer impedance 6 % · Every allowance in the load list (lighting, HVAC, heat tracing, auxiliaries, sockets, hoists) · UPS loads (VA), 2 × UPS, 30 min battery · Hydraulic estimate: benzene 857 kg/m³, P-100 Δp 35 bar, pump η 0.60–0.75, gas k 1.4, compressor η 0.75.